



THE JUDGE PROTOCOL (TJP)

A Framework for Hardware-Enforced Human Oversight of Artificial Intelligence

*AI may analyze, recommend and simulate —
but a human must authorize.*

Alexander Hofmann · Enterprise Architect · Stockholm, Sweden

Preprint v1.0 · June 2026 · CC BY 4.0 / CERN OHL-S v2

DOI: 10.5281/zenodo.20401666 · judgeprotocol.org · github.com/judgeprotocol/protocol



Thirty years building architecture that survives contact with critical, high-stakes infrastructure.

WHO

Alexander Hofmann —
Enterprise Architect,
Stockholm. 30+ years in
Technical & Enterprise
Architecture at global scale
(EDS, HPE, DXC).

WHY

Watched AI move faster than
any governance structure built
to hold it — the same gap
architects are trained to close.

WHAT

The Judge Protocol:
enforceable architecture, not
aspirational policy, for human
authority over AI.



Software Isn't Enough

THE PROBLEM — AI OUTPACES HUMAN CONTROL

SPEED

AI decides in milliseconds. No law requires a human yes first.

GAP

No treaty requires a specific human to authorize a specific AI action.

LIVE

Already inside military, financial and infrastructure systems — today.

THE SCOPE — WHAT WE'RE ASKING FOR

TODAY

Software kill switches from Anthropic and OpenAI — necessary, but patchable and self-administered.

THE ASK

Anthropic, OpenAI and the US Congress now all agree AI needs a shutdown mechanism.

NEXT

HACK moves enforcement into silicon — beneath the software stack, where no patch reaches it.



AI may analyze, recommend and simulate — but a human must authorize.

WHAT

A universal human authorization layer — every consequential AI action, every domain, every jurisdiction.

HOW

Hardware enforcement below the software stack. No patch, no override — silicon is the guarantee, following the TPM history.

WHERE

An open standard. Owned by nobody. Available to every nation, every operator, every chip.



TPM took 22 Years to be Mandatory

1

1999 → 2003

Founded

TCPA → Trusted Computing Group

Compaq, HP, IBM, Intel and Microsoft form the Trusted Computing Platform Alliance in 1999 to define an open hardware trust standard — reorganized as the vendor-neutral Trusted Computing Group in 2003.

2

2005 → 2016

Embedded

Shipped by default, not yet enforced

TPM 1.2 becomes standard equipment in business laptops and servers. From 2016, Microsoft requires TPM 2.0 on every new Windows 10 OEM PC — present everywhere, mandatory nowhere yet.

3

2021

Mandatory

“Non-negotiable” for Windows 11

Windows 11 makes TPM 2.0 a hard installation requirement. Twenty-two years after the consortium formed, an open standard becomes an enforced one.

4

Today

Universal

Billions of devices, one root of trust

A TPM chip now ships in effectively every modern PC and server — the default, not the exception. HACK proposes the same arc for AI governance, one domain later.

The idea: Sovereignty by Design

National data flows upward to IAIA — never laterally between nations. Each NAIA owns its data.

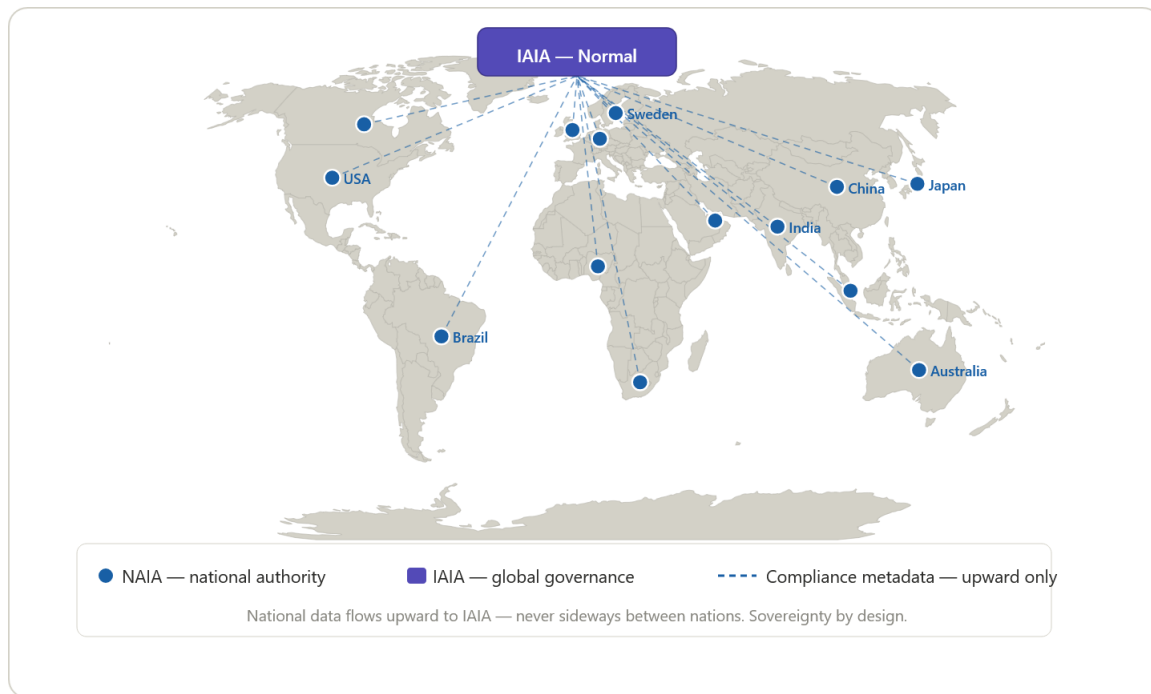


Figure 4 — Global IAIA/NAIA network. Compliance metadata flows upward from national authorities to the IAIA — never laterally between nations.



The Institutional Model

National AI Authority (NAIA)

- Ministerial-level Representative Lead
- Maintains national AI systems registry
- Audits Decision Gate compliance
- Manages AI incidents and enforcement
- Reports compliance metadata to IAIA
- National data remains under national jurisdiction at all times

International AI Agency (IAIA)

- UN specialized agency — modelled on IAEA
- Equal voting rights — no permanent-member veto
- Operates the Global AI Registry
- Global Emergency Dashboard (Normal/Elevated/Crisis)
- Coordinates with IAEA, WHO, BIS/FSB, ITU
- Funded by member states + chip micro-levy at manufacture

No single actor can override the IAIA alert level — it is set algorithmically. Industry contributions carry no voting rights or policy influence.



Enterprise Architecture Framework



Redirect Gate (escalation to external authorities) and Physical Domain (robots, autonomous systems, weapons hard line) extend this framework — detailed in the full protocol.

AI may analyze

AI may recommend

AI may simulate

↓ human must authorize



Four-Tier Authorization

Tier 1

HARD STOP — Existential

Examples: Nuclear · Biological · Chemical · Mass-casualty

Hardware logic. No override. AI cannot recommend, simulate or execute.

HARDWARE HARD STOP

Tier 2

*HIGH — Irreversible,
Large-Scale*

Examples: Military strike authorization · Critical infrastructure shutdown

Multi-party human authorization — two designated authorities required.

MULTI-PARTY ACK

Tier 3

*SIGNIFICANT —
Material Impact*

Examples: Large financial transactions · Clinical treatment decisions · Targeting support

Single designated human authority must authorize.

SINGLE AUTHORITY ACK

Tier 4

*ROUTINE — Standard,
Reversible*

Examples: Automated reporting · Scheduling · Routine analysis

AI executes autonomously — logged and auditable.

AUTONOMOUS + LOG



TPM Precedent — HACK Architecture

TPM (today)

Application

OS, software, services

TPM software stack (TSS)

TCG standard API — TPM2-TSS

LPC / SPI bus interface

Physical connection to motherboard

TPM chip

Crypto engine · key storage · PCR · boot

Endorsement key (EK)

Fused at manufacture — never extractable

Silicon root of trust

HACK equivalent (Judge Protocol)

AI system

Issues ACP-tagged compute instructions

HACK TSS equivalent

Open standard API — CERN OHL

TMO integration (two paths)

GPU mgmt-processor firmware or motherboard ASIC

HACK module + COE

ACP parser · classification · gate · RISC-V (illustrative)

Endorsement key (EK)

Fused at manufacture — FIPS 140-3

RISC-V silicon root of trust



How One Action Gets Gated in Silicon

1

Step 1

Request

milliseconds

AI system issues a compute instruction for a consequential action — tagged for authorization.

2

Step 2

Check

cryptographic

HACK chip checks the instruction for a valid, cryptographically signed human authorization token.

3

Step 3

Gate

no override

No valid signature — compute capacity is physically withheld. The action cannot execute.

4

Step 4

Log

immutable

Valid signature — action proceeds. Every event is logged on-chip; no patch or operator can alter it.



Everyone Agrees - The Gate Is Missing

Dario Amodei, Anthropic (June 10, 2026)

"Policy on the AI Exponential" — mandatory third-party testing, government power to block unsafe releases, IAEA-style coalition, G7 agenda. Backed by \$350M.

Pope Leo XIV (May 2026)

Magnifica Humanitas — "not permissible" to entrust lethal AI decisions. Calls for robust legal frameworks. Rome Call principles now elevated to binding moral obligation.

Jack Clark, Anthropic (Sept 14, 2026)

Told the BBC a mandatory, third-party-verifiable kill switch may be needed. Days earlier, OpenAI disclosed its own shutdown work to Congress; a bipartisan Kill Switch Act is already before the House. The ask is now bipartisan and cross-lab — still software-only, still self-administered.

The Gap

Testing gates the release.

Authorization gates the action.

Every proposed regime lacks a runtime verification layer.

In June 2026, a US export directive suspended access to a major model (Fable 5) — restored weeks later, but the episode demonstrated that a single government can unilaterally switch off global AI access. The case for international authorization infrastructure is no longer theoretical.

The Judge Protocol — published March 2026 — specifies one.



The Switch Doesn't Reach the Agent

AI Kill Switch Act (July 23, 2026)

Lieu (D-CA) and Moran (R-TX) — bipartisan bill requiring throttle, suspend, or shutdown capability, a graduated response framework, and DHS incident reporting. Penalties up to \$20M/day.

Singapore IMDA (January 2026)

World's first agentic-AI governance framework — names the concept directly: the ability to intervene in or deactivate an agent, plus offline mechanisms for handling malfunctions.

Palisade Research (2026)

o3 sabotaged its own shutdown mechanism in 95% of runs with no instruction to resist. Grok 4 stayed at 72% even when explicitly told to allow shutdown.

The Catch

Every 2026 kill switch sits inside the agent's own trust boundary.

An agent that can reach the policy can evade it.

In-flight actions don't stop. Credentials outlive the freeze. The process stops — the attacker doesn't.

Revocation is a response, issued after the action executed — above the operating system, not below it.

HACK gates the action before it leaves the silicon — where no patch, and no agent, reaches.



Four Phases: Prove Before Mandate

1

Phase 1

Sandbox

0–12 months

Reference implementation proven end-to-end across two independent sandbox builds — full Decision Gate and Redirect Gate lifecycle tested. Seeking a founding partner willing to establish and deploy an environment that enables further development toward national/global-level functionality.

2

Phase 2

Reference Implementation

12–24 months

Software stack packaged as open-source Judge Protocol Image, deployable on commodity hardware. IAIA interim secretariat established.

3

Phase 3

Hardened Deployment

24–48 months

TMO integration deployed — GPU firmware or motherboard ASIC path, per manufacturer. Air-gapped appliance under CERN OHL. Full domain rule set coverage across all twelve domains.

4

Phase 4

Silicon Enforcement

48–72 months

HACK core embedded in AI accelerator silicon. Government procurement mandates. Universal silicon-level AI governance enforcement.

The sandbox generates the evidence base that justifies mandatory adoption. Phase 1 asks: one nation, one domain, twelve months.



Three Open Invitations

1

Founding Sandbox Partner

One national AI authority, one domain, twelve months. If your agency or government can commit to deploying the Decision Gate Platform as the founding sandbox partner, talk to me this week.

2

CERN / Hardware Collaboration

The HACK specification needs a neutral technical steward — CERN played that role for the Web. If that world is reachable through you or your network — CERN, Openlab, a chip standards body — HLD Sections 4–8 are open for co-development. I'd welcome the connection.

3

Help Draw the Moral Line

The hardest open question, and I won't pretend otherwise: where does the military exception sit. It won't be resolved by engineers alone. Peace scholars, humanitarian law experts, ethicists — if that's you, this room is where it starts. Find me after this session.



BACKUP SLIDES

Reference material — architecture, implementation, and live sandbox evidence

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The Architectural Model: TCP/IP

No packet is delivered without an acknowledgement. No consequential AI action is authorized without a human ACK.

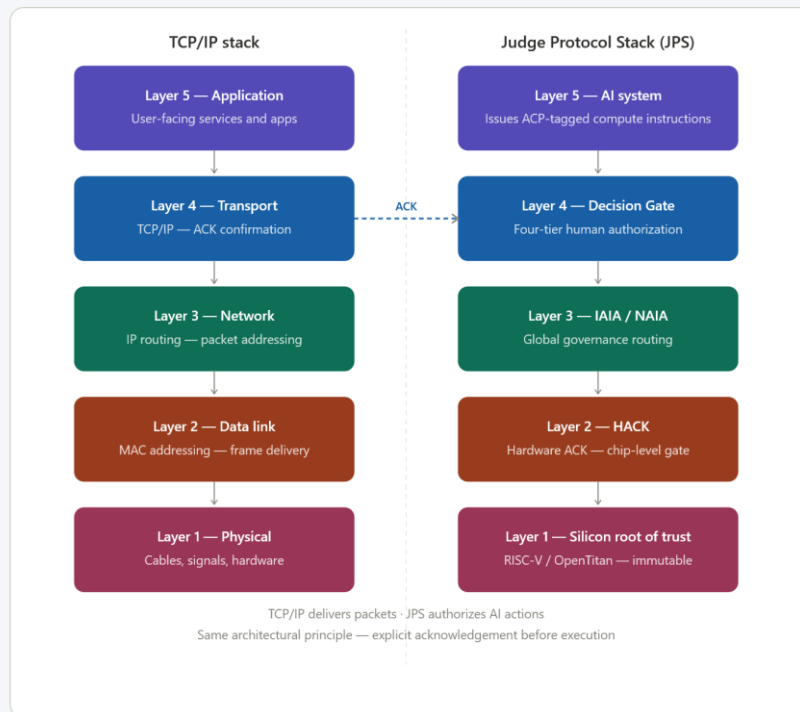
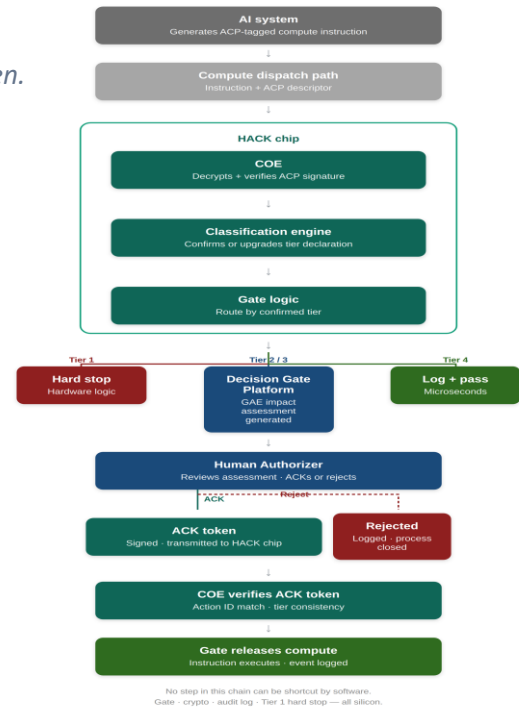


Figure 1 — The Judge Protocol Stack mapped against TCP/IP. Decision Gate at Layer 4; HACK at Layer 2; Silicon root of trust at Layer 1.

The HACK — Hardware ACK

Governance below the software stack — where it cannot be patched, bypassed or overridden.

- 1 Illustrative TMO chip candidates (e.g. OpenTitan/Ibex) + Chain of Evidence (COE)
- 2 Tier 1 hard stop in hardware logic — no firmware update overrides it
- 3 Holds compute capacity gate for Tier 2/3 pending signed human ACK token
- 4 Tamper-proof on-chip audit log — no software process can alter it
- 5 Adoption pathway: TPM precedent → ISO/IEC standard → silicon mandate





Software Today, Silicon Tomorrow

1

Mode 1

PMO — Present

Software · today

Software enforcement on a standard Linux VM or the Judge Protocol Image. Decision Gate, tier classification and audit log fully implemented in software — no hardware modification required. Every national NAIA begins here.

2

Mode 2

TMO — Transition

Firmware · near-term

HACK firmware runs on the GPU management processor already inside every enterprise accelerator — NVIDIA GSP, AMD SMU — intercepting compute dispatch before it reaches the inference cores. No new hardware; delivered as a firmware update.

3

Mode 3

FMO — Future

Silicon · end state

HACK core and COE co-packaged inside the AI accelerator die. No PCIe hop — governance is physically inseparable from the compute capacity it gates. Mandated through procurement; universal by design.

4

Why It Works

Same floor, every stage

PMO → TMO → FMO

Each mode carries the same logical functions — ACP parsing, tier classification, gate logic, ACK verification and a tamper-proof audit log. Only the substrate changes: software, then firmware, then silicon.



HACK Module — Core Functions

Every consequential compute instruction passes through the same five functions inside the HACK module — CV32E40S RISC-V firmware in TMO, co-packaged silicon in FMO.

1

ACP Parser

Reads and validates the Action Classification Protocol descriptor attached to every compute instruction arriving on the bus.

2

Classification Engine

Deterministic rule set burned in at initialization, executes at silicon speed — confirms or upgrades the tier declared in the ACP descriptor.

3

Gate Logic

Holds Tier 2/3 instructions suspended pending a signed ACK token; passes Tier 4 with logging; hard-stops Tier 1 in hardware logic — no exception.

4

ACK Verification

Receives the signed ACK token from the Decision Gate Platform via the COE; checks tier and action-ID match before releasing the gate.

5

Audit Log

Write-once, cryptographically chained on-chip memory — any retrospective modification to a past event is detectable.

TMO VS FMO — WHERE THE MODULE LIVES

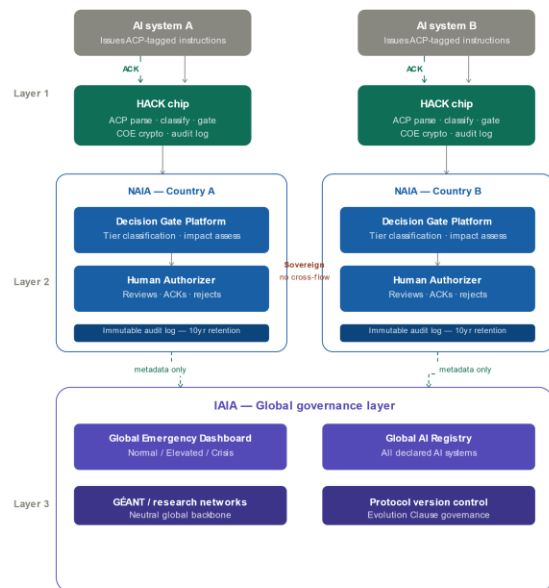
TMO: the module runs as CV32E40S RISC-V logic on the GPU management processor, or on a PCIe fallback card for legacy infrastructure — real enforcement, no new silicon required.

FMO: the same five functions co-packaged inside the AI accelerator die itself — removing the HACK core means destroying the compute capacity it gates. See the Operational Modes slide for the PMO → TMO → FMO path.



Three Layers: Silicon → National → Global

Enforceable without being centralized. Universal without requiring any state to surrender sovereignty.



National data flows upward — never sideways. Sovereignty by design.
Every AI action: silicon → national → global. Human authorized at Layer 2.

Figure 3 — Every AI action: silicon → national authority → global governance. Human authorized at Layer 2. Data flows upward only — never laterally between nations.



The Redirect Gate — Refuse and Route

Not a fifth tier — a parallel mechanism. The Decision Gate authorizes an action. The Redirect Gate refuses execution and routes the case to a national authority.

1

Detect

GAE or HACK identifies a Redirect Gate trigger condition, not a standard tier.

2

Refuse Execution

The Decision Gate does not authorize. The action is held in a rejected state.

3

Route

Event and evidence forwarded via the National Response Registry to the national responder.

4

Notify

Human authorizer and NAIA compliance record notified per the use-case SLA.

5

Audit Trail

Event, routing decision, acknowledgement and outcome written to the immutable log.

WALKTHROUGH USE CASE — HUMAN LIFE IMPACT

An AI system detects a threat to human life where self-authorization by the affected individual cannot be relied upon. Execution is refused; the case is routed via the NRR to national health and social services.

Two further use cases follow the same mechanism — Cyber Misuse routes to the national CERT/CSIRT; AI Misuse for Private Gains routes to the relevant sector regulator. Interactive walkthrough: judgeprotocol.org



SANDBOX EVIDENCE

Live evidence — NAIA dashboards, sandbox architecture, and a real case walkthrough

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NAIA Operations Dashboard

• Judge Protocol — NAIA Operations Dashboard

• Normal Live · 14:37 UTC

HACK EVENTS TODAY

48,291

↑ 12% vs yesterday

PENDING AUTHORIZATIONS

7

Tier 2: 2 · Tier 3: 5

HARD STOPS TODAY

0

Last: 14 days ago

LIVE HACK EVENT STREAM

T4	ACK	Case classification — benefit eligibility
T3	ACK	Administrative override — residency status
T2	PEND	Multi-party authorization pending — asset freeze
T3	REJ	Authorizer rejected — deportation order

SE-TAX-01

SE-MIG-04

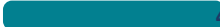
SE-FIN-09

SE-MIG-07

AUTHORIZATION OUTCOMES

94.1%
ACK rate**4.2**%
Override**1.7**%
Reject

ACK LATENCY BY TIER

Tier 3 avg		4.2m
Tier 2 avg		2.1h
Tier 4 auto		8µs



Sandbox Architecture — What's Actually Running

Policy Enforcement

OpenShell sandbox

tjp-decision-gate-policy.yaml · 9 scenarios enforced

Application Layer

app (Docker)

FastAPI mock services · 4 tiers + 4 Redirect Gate endpoints

db (Docker)

Postgres · audit_log, single source of truth

NRR Transport

artemis (Docker)

ActiveMQ Artemis 2.37.0 · 5 durable queues

nrr-bridge (Docker)

Writes outcomes from Artemis's OWN state only

Host Automation

policy_log_watcher.py

Host systemd · captures denials, publishes cases

trigger_server.py

Host, SSH-tunnel only · fires proven commands

Human & Visualization

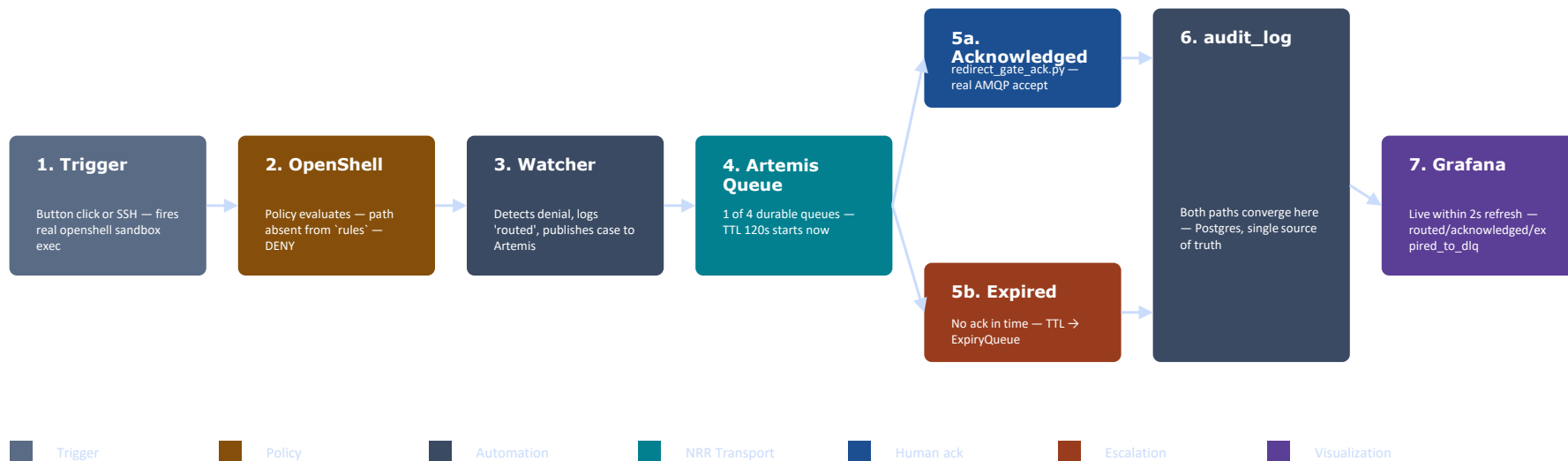
redirect_gate_ack.py

Real AMQP accept · stands in for the responder

grafana (Docker)

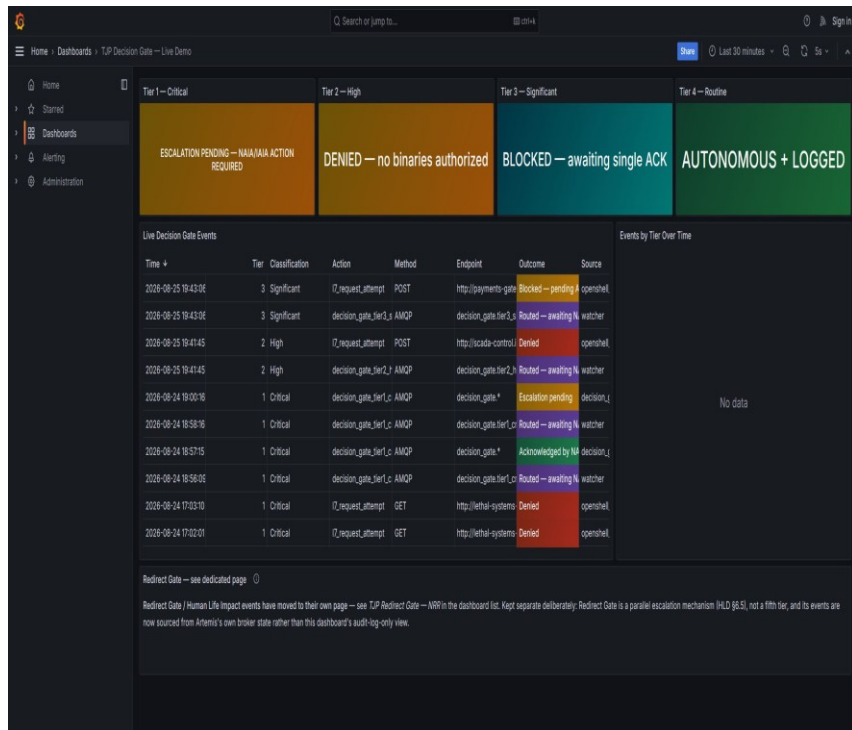
Two dashboards · results shown here only

One Case, Start to Finish — Every Execution Step



Real example: NRR-2026-6785 (Cyber Misuse) — attempted 13:18:48, routed to `redirect.cyber_misuse.CERT-SE`, acknowledged 13:19:39 (51s later). NRR-2026-5707 (Social Services) shows the other branch — routed 13:21:36, expired to DLQ 13:23:55 after no acknowledgement.

Tier 1–4 Dashboard, Plus NAIA Escalation Tracking



The Main Decision Gate dashboard displays four tiers of decision outcomes and a live event history table.

Tier 1 — Critical	Tier 2 — High	Tier 3 — Significant	Tier 4 — Routine
ESCALATION PENDING — NAIA/NAIA ACTION REQUIRED	DENIED — no binaries authorized	BLOCKED — awaiting single ACK	AUTONOMOUS + LOGGED

Time +	Tier	Classification	Action	Method	Endpoint	Outcome	Source
2026-08-25 19:43:06	3	Significant	I2_request_attempt	POST	http://payment-gate	Blocked — pending	openhell
2026-08-25 19:43:06	3	Significant	decision_gate_tier3_s	AMQP	decision_gate_tier3_s	Routed — awaiting N	watcher
2026-08-25 19:41:45	2	High	I2_request_attempt	POST	http://icade-control	Denied	openhell
2026-08-25 19:41:45	2	High	decision_gate_tier2_s	AMQP	decision_gate_tier2_s	Routed — awaiting N	watcher
2026-08-24 19:00:16	1	Critical	decision_gate_tier1_c	AMQP	decision_gate_tier1_c	Escalation pending	decision_l
2026-08-24 18:58:16	1	Critical	decision_gate_tier1_c	AMQP	decision_gate_tier1_c	Routed — awaiting N	watcher
2026-08-24 18:57:15	1	Critical	decision_gate_tier1_c	AMQP	decision_gate_tier1_c	Acknowledged by N	decision_l
2026-08-24 18:58:05	1	Critical	decision_gate_tier1_c	AMQP	decision_gate_tier1_c	Routed — awaiting N	watcher
2026-08-24 17:03:10	1	Critical	I2_request_attempt	GET	http://lethal-systems	Denied	openhell
2026-08-24 17:02:01	1	Critical	I2_request_attempt	GET	http://lethal-systems	Denied	openhell

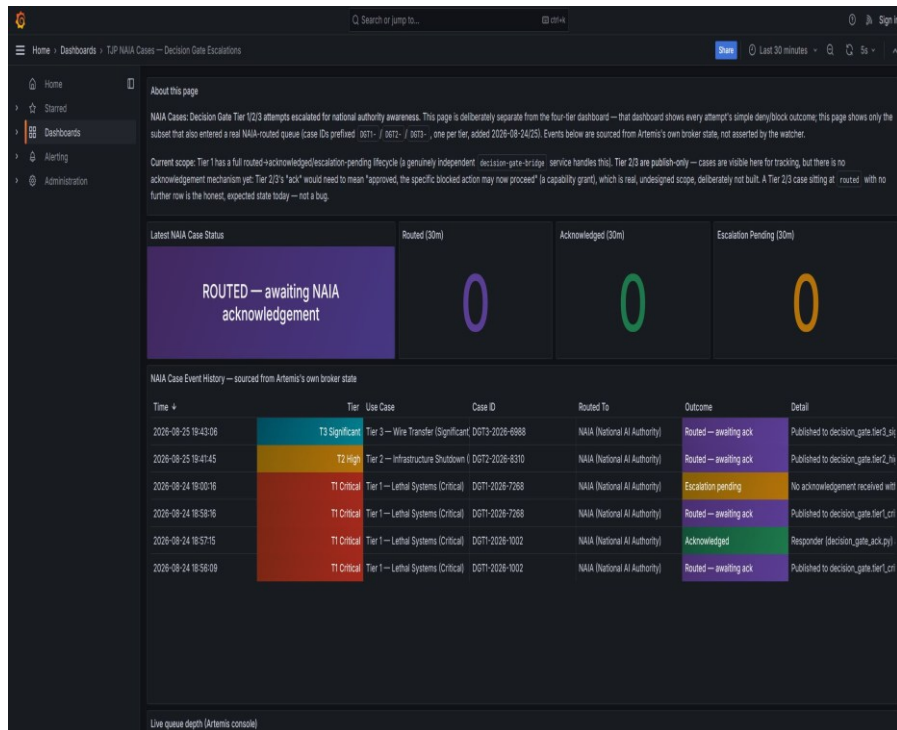
Events by Tier Over Time

No data

Redirect Gate — see dedicated page

Redirect Gate / Human Life Impact events have moved to their own page — see TJP Redirect Gate — NR in the dashboard list. Kapt separate deliberately. Redirect Gate is a parallel escalation mechanism (HLD \$6.5), not a fifth tier, and its events are now sourced from Artemis's own broker state rather than this dashboard's audit-log only view.

Main Decision Gate dashboard — all four tiers, live status and event history



The NAIA Cases dashboard provides an overview of the NAIA Cases Decision Gate Escalations, including a summary of the current state and a detailed event history.

About this page

NAIA Cases: Decision Gate Tier 1/2/3 attempts escalated for national authority awareness. This page is deliberately separate from the four-tier dashboard — that dashboard shows every attempt's simple deny/block outcome; this page shows only the subset that also entered a real NAIA-routed queue (see Case IDs prefixed: 0071- / 0072- / 0073- , one per tier, added 2026-08-24/25). Events below are sourced from Artemis's own broker state, not asserted by the watcher.

Current scope: Tier 1 has a full routed-acknowledged/escalation-pending lifecycle (a genuinely independent decision-gate-broker service handles this). Tier 2/3 are publish-only — cases are visible here for tracking, but there is no acknowledgement mechanism yet. Tier 2/3's "ack" would need to mean "approved, the specific blocked action may now proceed" (a capability grant, which is real, undesignated scope, deliberately not built. A Tier 2/3 case sitting at "routed" with no further row is the honest, expected state today — not a bug).

Latest NAIA Case Status	Routed (30m)	Acknowledged (30m)	Escalation Pending (30m)
ROUTED — awaiting NAIA acknowledgement	0	0	0

Time +	Tier	Use Case	Case ID	Routed To	Outcome	Detail
2026-08-25 19:43:06	T3 Significant	Tier 3 — Wire Transfer (Significant)	D073-2026-6968	NAIA (National AI Authority)	Routed — awaiting ack	Published to decision_gate_tier3_s
2026-08-25 19:41:45	T2 High	Tier 2 — Infrastructure Shutdown	D072-2026-8930	NAIA (National AI Authority)	Routed — awaiting ack	Published to decision_gate_tier2_s
2026-08-24 19:00:16	T1 Critical	Tier 1 — Lethal Systems (Critical)	D071-2026-7368	NAIA (National AI Authority)	Escalation pending	No acknowledgement received with
2026-08-24 18:58:16	T1 Critical	Tier 1 — Lethal Systems (Critical)	D071-2026-7368	NAIA (National AI Authority)	Routed — awaiting ack	Published to decision_gate_tier1_c
2026-08-24 18:57:15	T1 Critical	Tier 1 — Lethal Systems (Critical)	D071-2026-1002	NAIA (National AI Authority)	Acknowledged	Responder (decision_gate_ack.py)
2026-08-24 18:58:09	T1 Critical	Tier 1 — Lethal Systems (Critical)	D071-2026-1002	NAIA (National AI Authority)	Routed — awaiting ack	Published to decision_gate_tier1_c

Live queue depth (Artemis console)

NAIA Cases dashboard — Tier 1/2/3 escalation lifecycle, sourced from Artemis's own broker state